

Changyeon Kim

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Research Interest

My research centers on training intelligent robotic agents that align closely with human intentions. To achieve this, I leverage Reinforcement Learning (RL) to develop robotic agents capable of learning human-aligned behaviors in various tasks. A key focus of my work has been designing reward learning algorithms that accurately capture human intent, either through direct human feedback or by grounding in vision-language foundation models. More recently, I have emphasized advancing RL methods themselves to better support the training of such human-aligned robotic agents.

Education

Korea Advanced Institute of Science and Technology

M.S. + PH.D. IN ARTIFICIAL INTELLIGENCE

Advisor: Kimin Lee and Jinwoo Shin

Daejeon, S.Korea

Feb. 2022 - Present

Korea Advanced Institute of Science and Technology

B.SC. IN COMPUTER SCIENCE AND MATHEMATICS (MINOR)

Daejeon, S.Korea

Mar. 2016 - Feb. 2021

Work Experience

Visiting PhD Student @ University of Texas at Austin

Austin, USA

WITH YUKE ZHU

Sep. 2024 - Jun. 2025

- Developing an action-sequence reinforcement learning framework suited for training with modern action prediction architectures. [P2]
- Designed VLM-based reward framework encouraging human-aligned behaviors without human reward engineering.

External Collaborator

WITH JOSEPH J. LIM (KAIST)

Mar. 2024 - June. 2024

- Developed a visual reward learning algorithm [C4] for solving complex long-horizon robotic manipulations.

External Collaborator

WITH LISA LEE (GOOGLE DEEPMIND)

Apr. 2023 - Aug. 2023

- Developed an imitation learning algorithm [C3] using multimodal representations for improving generalization ability in unseen variations.

External Collaborator

WITH HONGLAK LEE (UNIVERSITY OF MICHIGAN)

Mar. 2022 - June. 2024

- Developed an imitation learning algorithm [C3] using multimodal representations for improving generalization ability in unseen variations.
- Developed a reinforcement learning algorithm [W1] for improving generalization ability in varying dynamics.
- Developed a preference-based reinforcement learning algorithm [C2] for modeling non-Markovian human preferences.
- Developed a visual reward learning algorithm [C4] for solving complex long-horizon robotic manipulations.

Machine Learning Engineer

Seongnam, S.Korea

KAKAO, RECOMMENDATION TEAM

Dec. 2020 - Feb. 2022

- Developed ML platform for recommendation system.
- Implemented data pipeline from user feedback to refined user-item interaction matrix data.

Publications

C: Conference, W: Workshop, P: Preprint, *: Equal contribution

[P2] DEAS: DETached value learning with Action Sequence for Scalable Offline RL

CHANGYEON KIM, HAEONE LEE, YOUNGGYO SEO, KIMIN LEE, YUKE ZHU

- Preprint.
- Available upon request

[P3] HAMLET: Switch Your Vision-Language-Action Model into a History-Aware Policy

MYUNGKYU KOO, DAEWON CHOI, TAEYOUNG KIM, KYUNGMIN LEE, CHANGYEON KIM, YOUNGGYO SEO, JINWOO SHIN

- Preprint.
- Available upon request

[P4] Contrastive Representation Regularization for Vision-Language-Action Models

TAEYOUNG KIM, JIMIN LEE, MYUNGKYU KOO, DONGYOUNG KIM, KYUNGMIN LEE, **CHANGYEON KIM**, YOUNGGYO SEO, JINWOO SHIN

- Preprint.
- Available upon request

[P5] Verifier-free Test-Time Sampling for Vision Language Action Models

SUHYEOK JANG, DONGYOUNG KIM, **CHANGYEON KIM**, YOUNGSUK KIM, JINWOO SHIN

- Preprint.
- Available upon request

[C5] B-MoCA: Benchmarking Mobile Device Control Agents across Diverse Configurations

Philadelphia, USA

JUYONG LEE, TAYWON MIN, MINYONG AHN, DONGYOON HAHM, HAEONE LEE, **CHANGYEON KIM**, KIMIN LEE

Aug, 2025.

- Conference on Lifelong Learning Agents (CoLLAs), 2025.
- ICLR 2024 Workshop on Generative Models for Decision Making (GenAI4DM), **Spotlight**

[C4] Subtask-Aware Visual Reward Learning from Segmented Demonstrations

Singapore, Singapore

CHANGYEON KIM, MINHO HEO, DOOHYUN LEE, JINWOO SHIN, HONGLAK LEE, KIMIN LEE, JOSEPH J LIM

Apr, 2025.

- International Conference on Learning Representations (ICLR), 2025.

[C3] Guide Your Agent with Adaptive Multimodal Rewards

New Orleans, USA

CHANGYEON KIM, YOUNGGYO SEO, HAO LIU, LISA LEE, JINWOO SHIN, HONGLAK LEE, KIMIN LEE

Dec, 2023.

- Conference on Neural Information Processing Systems (NeurIPS), 2023.
- **Finalist**, Qualcomm Innovation Fellowship Korea 2024

[C2] Preference Transformer: Modeling Human Preferences using Transformers for RL

Kigali, Rwanda

CHANGYEON KIM*, JONGJIN PARK*, JINWOO SHIN, HONGLAK LEE, PIETER ABBEEL, KIMIN LEE

May, 2023.

- International Conference on Learning Representations (ICLR), 2023.

[W1] Dynamics-Augmented Decision Transformer for Offline Dynamics Generalization

New Orleans, USA

CHANGYEON KIM*, JUNSU KIM*, YOUNGGYO SEO, KIMIN LEE, HONGLAK LEE, JINWOO SHIN

Nov, 2022.

- NeurIPS 2022 Workshop on Offline Reinforcement Learning (NeurIPSW)

[C1] Collecting the Public Perception of AI and Robot Rights

Online

GABRIEL LIMA, **CHANGYEON KIM**, SEUNGHO RYU, CHIHYOUNG JEON, MEEYOUNG CHA

Oct, 2020.

- Conference on Computer-Supported Cooperative Work and Social Computing (CSCW), 2020.

[P1] MOI-Mixer: Improving MLP-Mixer with Multi Order Interactions in Sequential Recommendation

HOJOON LEE, DONGYOON HWANG, SUNGHWAN HONG, **CHANGYEON KIM**, SEUNGRYONG KIM, JAEGUL CHOO

- ArXiv Preprint.

Honors & Awards

2024	Finalist , Qualcomm Innovation Fellowship Korea	Seoul, S.Korea
2023	Travel Award , Conference on Neural Information Processing Systems (NeurIPS)	New Orleans, USA
2023	Scholarship , KAIST-Google Partnership Program	Daejeon, S.Korea
2023	East Asia Student Travel Grant , Google	New Orleans, USA
2023	Travel Award , International Conference on Learning Representations (ICLR)	Kigali, Rwanda
2019	Dean's List (Fall Semester) , Department of Engineering, KAIST	Daejeon, S.Korea
2019	Line Scholarship (Fall Semester) , School of Computing, KAIST	Daejeon, S.Korea
2017 - 19	National Science and Engineering Scholarship , Korea Ministry of Science and ICT	Daejeon, S.Korea
2017	Kwanjeong Scholarship (Spring Semester) , KAIST	Daejeon, S.Korea

Invited Talks

Guide Your Agent with Adaptive Multimodal Rewards

NeurIPS 2023

IMITATION LEARNING FRAMEWORK WITH VLM REWARDS FOR BETTER GENERALIZATION

- LG AI Research, New Orleans, USA (2023)

Academic Services

Conference Reviewer ICML (2024, 2025), NeurIPS (2024, 2025), ICLR (2025, 2026), AAAI (2026)
Workshop Reviewer ICML Frontiers4LCD 2023, CoRL MRM-D 2024